

NAMMATUTOR GUESS PAPER

Electricity

Board-pattern practice paper, with full solutions

CLASS 12 · PHYSICS · CHAPTER 3

QUESTIONS

14

TOTAL
MARKS

35

TIME

75
minutes

PATTERN

CBSE

Current Electricity — Practice Paper

Class 12 · Physics · Chapter 3 · Board pattern

SAMPLE

Max Marks: 35

Time: 75 min

General instructions

1. This paper has five sections — A, B, C, D and E.
2. Section A carries 1 mark per question, Section B 2 marks, Section C 3 marks, Section D 5 marks.
3. Section E is a case-study question carrying 4 marks, with sub-parts.
4. Use standard values: $e = 1.6 \times 10^{-19}$ C where required. Draw circuit diagrams wherever asked.

SECTION A · 1 MARK EACH

Multiple choice — choose the correct option

1. The SI unit of resistivity is: [1]
(a) ohm (b) ohm-metre (c) siemens (d) ohm/metre
2. Two resistors of $6\ \Omega$ and $3\ \Omega$ are connected in parallel. Their equivalent resistance is: [1]
(a) $9\ \Omega$ (b) $4.5\ \Omega$ (c) $2\ \Omega$ (d) $1\ \Omega$
3. Kirchhoff's junction rule is a statement of conservation of: [1]
(a) energy (b) charge (c) momentum (d) mass
4. As temperature increases, the resistance of a pure metallic conductor: [1]
(a) decreases (b) increases (c) stays constant (d) becomes zero
5. A cell of EMF ϵ and internal resistance r is connected to an external resistance R . The terminal voltage is always: [1]
(a) $> \epsilon$ (b) $= \epsilon$ (c) $< \epsilon$ (d) zero

SECTION B · 2 MARKS EACH

6. Define drift velocity. Write the relation between current I and drift velocity v_d for a conductor of cross-section A carrying n free electrons per unit volume. [2]
7. Three resistors $2\ \Omega$, $3\ \Omega$ and $4\ \Omega$ are connected in series across a 9 V battery of negligible internal resistance. Find the current in the circuit. [2]
8. State the working principle of a potentiometer. Why is it preferred over a voltmeter to measure EMF? [2]

SECTION C · 3 MARKS EACH

9. Derive the expression for the equivalent resistance of three resistors R_1 , R_2 , R_3 connected in parallel. [3]
10. A battery of EMF 12 V and internal resistance $2\ \Omega$ is connected to a resistor of $4\ \Omega$. Find (a) the current in the circuit, (b) the terminal voltage of the battery. [3]
11. State Kirchhoff's two laws for electrical circuits. On what conservation principle is each law based? [3]

Current Electricity — Practice Paper (continued)

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SAMPLE

Sections D–E

SECTION D · 5 MARKS EACH

12. Derive an expression for the equivalent EMF and internal resistance when two cells of EMF ϵ_1 , ϵ_2 and internal resistances r_1 , r_2 are connected in parallel. [5]
13. Using Kirchhoff's laws, find the current through each branch of a circuit with two batteries (10 V, 2 V) and three resistors ($2\ \Omega$, $4\ \Omega$, $6\ \Omega$) arranged in a two-loop network. Draw the circuit and show all working. [5]

SECTION E · CASE STUDY · 4 MARKS

14. A Wheatstone bridge is a four-arm resistor network used to measure an unknown resistance precisely — far more precisely than an ordinary ammeter-voltmeter method — by balancing a galvanometer to read zero current. [4]

In a balanced Wheatstone bridge, arms P, Q, R and S carry no current through the galvanometer when $P/Q = R/S$. This is used in metre-bridge experiments in the school lab.

- (i) State the balance condition of a Wheatstone bridge. [1]
(ii) In a bridge, $P = 10\ \Omega$, $Q = 20\ \Omega$, $R = 15\ \Omega$. Find S for balance. [2]
(iii) Why is the bridge most sensitive when all four arms have comparable resistance? [1]

QUICK REFERENCE — FORMULAS USED IN THIS PAPER

What it finds	Formula
Ohm's law	$V = IR$
Resistors in series	$R_{eq} = R_1 + R_2 + \dots$
Resistors in parallel	$1/R_{eq} = 1/R_1 + 1/R_2 + \dots$
Terminal voltage of a cell	$V = \epsilon - Ir$
Wheatstone bridge balance	$P/Q = R/S$

SECTION A

1. (b) ohm-metre

$$\rho = RA/l \rightarrow \Omega \cdot m$$

3. (b) charge

$$\sum I(\text{in}) = \sum I(\text{out}) \text{ at a junction}$$

5. (c) $< \epsilon$

$$V = \epsilon - Ir, \text{ always less during discharge}$$

2. (c) 2Ω

$$1/R = 1/6 + 1/3 = 1/2 \rightarrow R = 2\Omega$$

4. (b) increases

More lattice vibration $\rightarrow$ more collisions

SECTION B

6. Drift velocity = average velocity gained by electrons against the field, between collisions. $I = nAev_d$ [1: definition, 1: formula]

7. $R = 2+3+4 = 9\Omega$. $I = V/R = 9/9 = 1 \text{ A}$

8. Potentiometer compares EMFs using a balance-point on a uniform wire, drawing zero current from the cell at balance — so it measures true EMF, unlike a voltmeter which always draws some current and reads terminal voltage, not EMF.

SECTION C

9. Same voltage V across each; total current $I = I_1 + I_2 + I_3 = V/R_1 + V/R_2 + V/R_3$. Since $I = V/R_{eq}$: $1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$

10. (a) $I = \epsilon/(R+r) = 12/6 = 2 \text{ A}$

(b) $V = \epsilon - Ir = 12 - 4 = 8 \text{ V}$

11. Junction rule ($\sum I = 0$) $\rightarrow$ conservation of charge. Loop rule ($\sum \Delta V = 0$) $\rightarrow$ conservation of energy.

SECTION D

12. Treating the parallel combination as a single cell: $\epsilon_{eq} = (\epsilon_1 r_2 + \epsilon_2 r_1)/(r_1 + r_2)$, $1/r_{eq} = 1/r_1 + 1/r_2$. Derived from equating terminal voltage expressions for both branches and applying charge conservation at the junction.

13. Apply loop rule to both loops and junction rule at the node; solve the resulting simultaneous equations for the three branch currents. (Full numeric working shown in the complete answer key, with the labelled circuit diagram.)

SECTION E – CASE STUDY

14.

(i) Balance condition: $P/Q = R/S$

(ii) $10/20 = 15/S \rightarrow S = 15 \times 20/10 = 30 \Omega$

(iii) Comparable arm values give the galvanometer deflection maximum sensitivity to small changes near balance — very unequal arms make the null point hard to detect precisely.